AI PROMPT AND RESPONSE APPENDIX

Course: BDA400 – Data Science Tools and Techniques
Assignment: Assignment 1 – Markdown Buddy
Project: Student Performance Data Analysis
AI Tool: ChatGPT (GPT-6)

AI Assistance Declaration

I used ChatGPT (GPT-6) to help plan and improve my GitHub README, organize my R Markdown documentation, generate formatting suggestions, and review Markdown syntax. I also used AI to suggest code examples and improve the clarity of my documentation.

I reviewed the AI-generated suggestions, compared them with my R script, and made the necessary changes. I also successfully knitted the R Markdown document into HTML and reviewed the results. All final calculations were executed and checked in RStudio.

This appendix documents the main AI prompts, summarizes the responses, and explains the improvements made during the assignment. The first two stages summarize recorded interactions, while Step 3 includes sample prompts from the assignment instructions and the corresponding documented work.

Step 1: Understanding README Structure

Interaction 1 – Seed Prompt

Prompt:

"Explain what sections a good GitHub README for an R data analysis project should include."

AI Response Summary:

ChatGPT suggested several important sections for a GitHub README, including project overview, objectives, dataset description, installation instructions, usage, results, dependencies, visualizations, and limitations.

It explained that a good README should help other users understand the project and how to run it.

Interaction 2 – Refinement Prompt

Prompt:

"Revise the sections list so it's concise and uses Markdown headers and bullet formatting."

AI Response Summary:

ChatGPT reorganized the suggested sections using Markdown headings and bullet points. This made the README structure easier to understand and follow.

The revised outline provided a simpler starting point for documenting the Student Performance Data Analysis project.

Interaction 3 – Critique and Validation Prompt

Prompt:

"Check the Markdown syntax for correctness and readability."

AI Response Summary:

ChatGPT reviewed the Markdown structure and recommended consistent headings, bullet points, and spacing.

It also suggested removing unnecessary repeated sections to make the README more organized.

Student Validation and Improvements:

I compared the suggested README structure with the GitHub repositories for dplyr and ggplot2.

I selected important sections such as Overview, Installation, and Usage. I also included a Cheat Sheet section to provide useful R commands.

Reference repositories:

https://github.com/tidyverse/dplyr

https://github.com/tidyverse/ggplot2

Step 2: Generating and Refining the README

Interaction 1 – Seed Prompt

Prompt:

"Here's a summary of my R project: the project I provided. Generate a professional README.md file using Markdown."

AI Response Summary:

ChatGPT generated an initial README for the Student Performance Data Analysis project.

The README explained the project's purpose and included sections such as Overview, Installation, Usage, Cheat Sheet, Reproducibility, and Status.

Some of the initial file references needed to be updated to match the final R project.

Interaction 2 – Refinement Prompt

Prompt:

"Add sections for Installation, Example Code, and License. Keep tone concise and professional."

AI Response Summary:

ChatGPT improved the README by adding the requested sections.

It included an example of R code for working with student data and provided installation instructions. It also included a License section without assuming that a license had already been selected.

The response made the README more useful for someone who wanted to understand or run the project.

Interaction 3 – Critique and Validation Prompt

Prompt:

"Review the Markdown for syntax errors and suggest 2 improvements for clarity."

AI Response Summary:

ChatGPT reviewed the README and suggested two main improvements:

1. Ensure filenames, repository paths, and dependencies match the actual project.
2. Explain the expected analysis results, visualizations, and output files more clearly.

These recommendations helped improve the accuracy and readability of the documentation.

Student Validation and Improvements:

I reviewed the README and updated it to match the final R script.

The project uses `set.seed(42)` to generate data for 50 fictional students. It calculates descriptive statistics and creates a scatter plot showing the relationship between study hours and exam scores.

I also checked that the README described the following output files correctly:

- `student_data.csv`
- `summary_statistics.csv`

- `study_hours_vs_exam_score.png`

These files are generated inside the `outputs` folder when the script runs.

Step 3: Documenting an R Script with R Markdown

This stage documents the R Markdown work using the sample prompts provided in the assignment instructions. The results below describe the completed document and its verification rather than unverified AI responses.

Interaction 1 – Seed Prompt Example

Prompt:

"Here's my R script description. Suggest Markdown formatting and code block examples."

Documented Outcome:

The R Markdown document explains the purpose of the `student_performance_analysis.R` script and describes its main inputs.

It includes R code examples demonstrating how the dataset is generated, how summary statistics are calculated, and how the scatter plot is produced.

Interaction 2 – Refinement Prompt Example

Prompt:

"Add syntax highlighting and improve section organization (e.g., # Purpose, ## Inputs, ## Outputs)."

Documented Outcome:

The R Markdown document uses organized headings, a table of contents, and syntax-highlighted R code chunks.

The document contains the following main sections:

- Purpose
- Inputs
- Method and Code Examples
- Outputs
- Interpretation and Limitations
- Conclusion
- AI Assistance Disclosure

The formatting makes the document easier to read and understand.

Interaction 3 – Critique and Validation Prompt Example

Prompt:

"Is the Markdown consistent with RMarkdown best practices?"

Documented Outcome:

The R Markdown document was successfully knitted into HTML.

I reviewed the generated output to confirm that the title, table of contents, headings, code examples, summary statistics, and scatter plot displayed correctly.

I also reviewed the AI Assistance Disclosure section to ensure that it accurately described the work completed.

Verification and Refinement Record

Step 1 – README Structure

I compared the AI-generated README suggestions with the dplyr and ggplot2 GitHub repositories. I selected the sections that were most relevant to my project and organized them using Markdown formatting.

Step 2 – README Development

I checked the README against the actual R script and updated the filenames, dependencies, code examples, and output descriptions. I also reviewed the Markdown headings, tables, and code blocks.

Step 3 – R Markdown Documentation

I successfully knitted the R Markdown document into HTML and checked the rendered document. I verified that the sections, R code, descriptive statistics, scatter plot, and AI disclosure were displayed correctly.

Final Reflection on AI Assistance

ChatGPT was useful for organizing the project documentation and suggesting improvements to the README and R Markdown structure.

However, I still needed to review the suggestions and ensure they matched the actual R script. Some initial documentation details required correction, particularly filenames and output descriptions.

This process showed me that AI can help improve documentation, but its suggestions still need to be reviewed and verified.

The project uses synthetic student data, and all final calculations and outputs were checked using RStudio. I am responsible for the accuracy and final submission of this assignment.